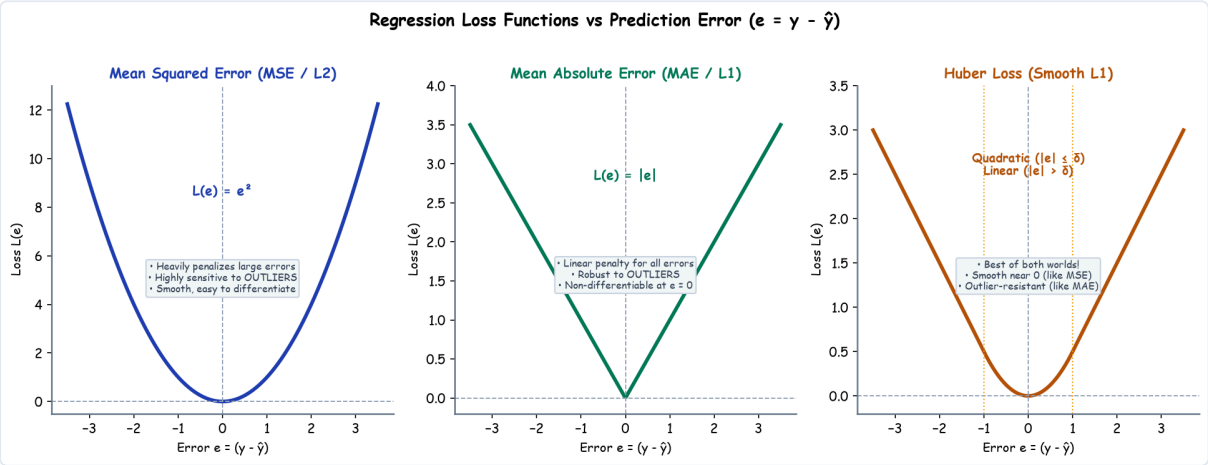
Submodule 37.02: Loss Functions (Regression)

1. Loss vs Cost: The Precise Technical Distinction

In deep learning literature and engineering discussions, the terms *Loss* and *Cost* are frequently conflated, yet represent distinct concepts:

- **Loss Function** $\mathcal{L}(y, \hat{y})$: Measures the prediction error on a **single isolated sample** (x_i, y_i).
- **Cost Function** $J(\theta)$: Measures the **aggregated average error** across the entire training dataset of N samples:

$$J(\theta) = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(y^{(i)}, \hat{y}^{(i)})$$



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2. Mean Squared Error (MSE / L2 Loss)

MSE is the classical standard loss function for continuous real-valued target estimation.

Mathematical Formulation

$$\mathcal{L}_{MSE}(y, \hat{y}) = (y - \hat{y})^2$$

$$J_{MSE}(\theta) = \frac{1}{2N} \sum_{i=1}^{N} (y^{(i)} - \hat{y}^{(i)})^2$$

(The scalar factor $\frac{1}{2}$ is intentionally introduced so that during derivative evaluation, the power of 2 cancels algebraically: $\frac{\partial}{\partial \hat{y}}[\frac{1}{2}(y-\hat{y})^2] = -(y-\hat{y})$).

Gradient Derivative

$$\frac{\partial J_{MSE}}{\partial \hat{y}} = -\frac{1}{N}\sum_{i=1}^{N}(y^{(i)} - \hat{y}^{(i)})$$

Advantages & Disadvantages

- **Advantages**:
    1. Strictly convex and smooth everywhere with a continuous gradient.
    2. Penalizes small errors gently while forcing the model to prioritize correcting large discrepancies.
- **Critical Disadvantage**:
    1. **Extreme Outlier Sensitivity**: Because errors are squared, an outlier with error $e = 100$ produces a penalty of $10,000$, disproportionately dominating parameter updates and skewing the regression boundary away from the true underlying distribution.

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3. Mean Absolute Error (MAE / L1 Loss)

MAE measures the linear distance between true and estimated targets without exponential amplification.

Mathematical Formulation

$$\mathcal{L}_{MAE}(y, \hat{y}) = |y - \hat{y}|$$

$$J_{MAE}(\theta) = \frac{1}{N}\sum_{i=1}^{N}|y^{(i)} - \hat{y}^{(i)}|$$

Gradient Derivative

$$\frac{\partial \mathcal{L}_{MAE}}{\partial \hat{y}} = \begin{cases} -1 & \text{if } y - \hat{y} > 0 \\ +1 & \text{if } y - \hat{y} < 0 \\ \text{undefined} & \text{if } y = \hat{y} \end{cases}$$

Advantages & Disadvantages

- **Advantages**:
    1. **Outlier Robustness**: Large errors only contribute linearly ($e = 100 \implies$ penalty 100).
- **Disadvantages**:
    1. **Non-smooth Derivative at Zero**: At $e = 0$, the function is non-differentiable (requires subgradient calculus).
    2. Constant gradient magnitude ± 1 causes updates to oscillate around the minimum rather than smoothly decelerating.

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4. Huber Loss (Smooth L1 Loss)

Huber Loss unifies the optimal attributes of both MSE and MAE by transitioning smoothly between quadratic and linear regimes governed by threshold hyperparameter δ:

Mathematical Formulation

$$\mathcal{L}_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta |y - \hat{y}| - \frac{1}{2}\delta^2 & \text{if } |y - \hat{y}| > \delta \end{cases}$$

Architectural Behavior

- For small residual errors ($|e| \leq \delta$): Becomes pure MSE $\implies$ smooth gradient deceleration to minimum.
- For large outlier errors ($|e| > \delta$): Becomes pure MAE $\implies$ linear scaling prevents outlier explosion.

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5. Comparative Trade-off Matrix

Metric Dimension	Mean Squared Error (MSE)	Mean Absolute Error (MAE)	Huber Loss (Smooth L1)
Mathematical Formula	$(y - \hat{y})^2$	$\lvert y - \hat{y} \rvert$	Piecewise Quadratic/Linear

Metric Dimension	Mean Squared Error (MSE)	Mean Absolute Error (MAE)	Huber Loss (Smooth L1)
Gradient at $e = 0$	Smoothly converges to 0	Discontinuous (± 1)	Smoothly converges to 0
Outlier Sensitivity	Severe (Quadratic explosion)	Low (Linear robustness)	Highly Robust (Linear beyond δ)
Computational Ease	$O(1)$ analytic differentiation	Requires subgradient handling	Piecewise conditional logic

| **PyTorch Equivalent** | `nn.MSELoss()` | `nn.L1Loss()` | `nn.HuberLoss(delta=1.0)` |\n